CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of B. Tech. Examination (CE/IT/EE)

December 2011

CL 103.01 Mechanics of Solids

Date: 12.12.2011, Monday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 Find the forces in all members of pin jointed truss loaded as shown in **Fig. 1.** **[07]**

Q - 2 (a) Define resultant and equilibrium. **[02]**

(b) Reduce the following S.I. units to units indicated. **[03]**

1000 MPa to kN/mm^2 10^9 kg to Gg 2000 μm to mm

(c) Two tensile forces of 20 kN and 30 kN are acting at a point with an angle of 60° between them. Find the magnitude and direction of the resultant force. **[03]**

(d) If resultant of a force system shown in **Fig. 2** is 15 kN vertical, find value of force P and α. **[06]**

OR

(d) Differentiate between moment and couple. Give characteristics of a couple. **[06]**

Q - 3 (a) A circle of 100 mm diameter is cut-out from circle of 500 mm diameter. The distance between centers of circle is 150 mm on left side to centre of big circle. Find centroid of the given lamina. **Refer Fig. 3.** **[07]**

(b) A ladder AB is shown in **Fig. 4.** The coefficient of friction between ladder and wall, and ladder and floor is 0.30. A man weighing 600 N is standing at the mid length of ladder. Calculate reaction at floor and wall. Neglect the weight of ladder. **[07]**

OR

Q - 3 (a) From a circular plate of 100 mm diameter a square plate of 50 mm diagonal is cutout as shown in **Fig. 5.** Find centroid of the remaining area. **[07]**

(b) A block weighing 150 kN is placed on a rough inclined plane making angle 30° with horizontal. If coefficient of friction is 0.25, find out the force to be applied on the block parallel to the plane so that the block is just on the point of moving up the plane. Also find angle of friction. **[07]**

SECTION – II

Q - 4 (a) Write Hook's law. **[02]**

(b) Enlist four mechanical properties of structural material and explain hardness test on metal. **[05]**

Q - 5 (a) Derive equation $\delta l = 4PL/\ \pi E d_1 d_2$ with usual notations. **[04]**

(b) A load of 2000 kN is applied on a short concrete column 500 mm x 500mm, reinforced with 4, 10mm dia. steel bars. Find stresses in concrete and steel. Take the value of E for steel as 2.1×10^5 N/mm^2 and for concrete 1.4×10^4 N/mm^2. **[05]**

(c) A steel bar 1600 mm long is acted upon by forces as shown in **Fig. 6**. Find the elongation of the bar. Take E = 210 GN/m^2. **[05]**

OR

Q - 5 (a) Explain stress-strain curve of mild steel. **[04]**

(b) A 200 mm long steel tube, 100 mm internal diameter and 10mm thick is surrounded by a brass tube of the same thickness and length. The composite section carries an axial compression of 100 kN. Find the load carried by each tube and shortening of each tube. Es = 0.2 MN/mm^2 , E_b = 0.1 MN/mm^2 **[05]**

(c) A steel bar 2m long and 200mm diameter is acted upon by 50 kN compression force. If E = 2×10^5 N/mm^2 and poisson's ratio is 0.25, find change in length and diameter. Also find linear strain & lateral strain. **[05]**

Q - 6 Draw shear force and bending moment diagrams indicating values at all important points for any two beams shown in **Fig. 7, Fig. 8 & Fig. 9.** Find out point of contra flexure if any. **[14]**

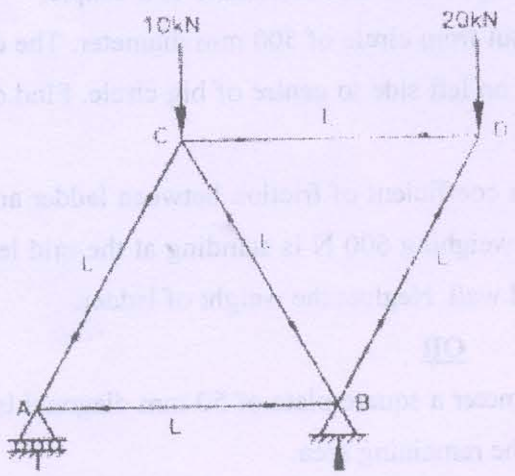

Fig. 1

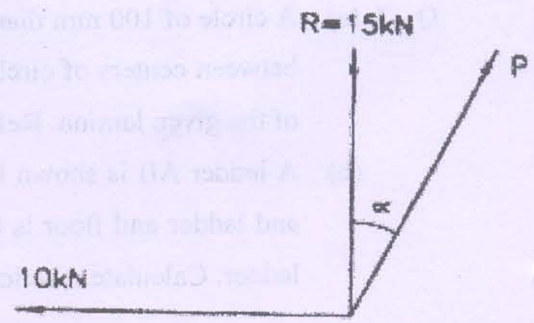

Fig. 2

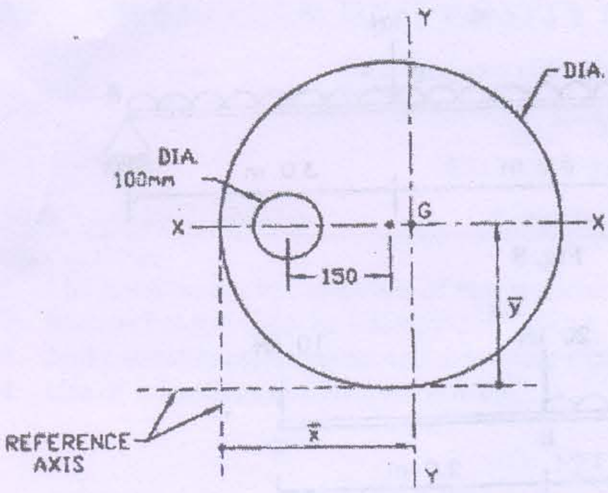

Fig. 3

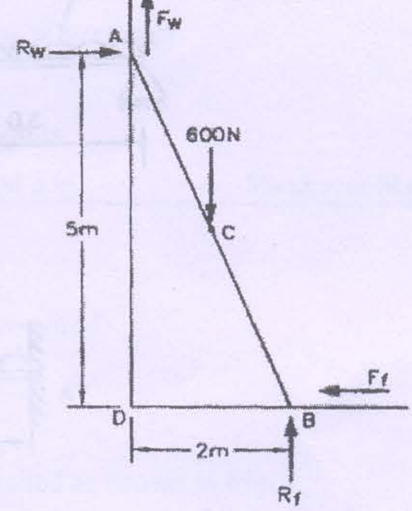

Fig. 4

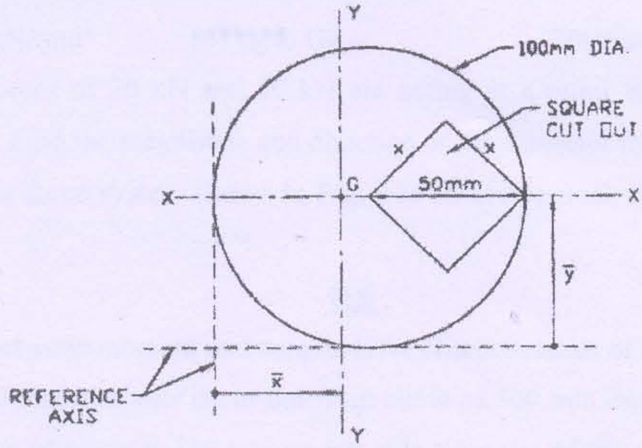

Fig. 5

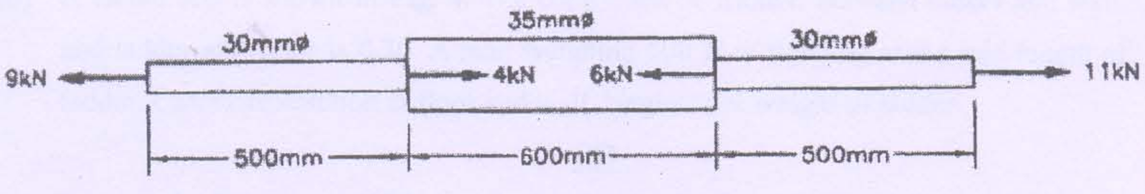

Fig. 6

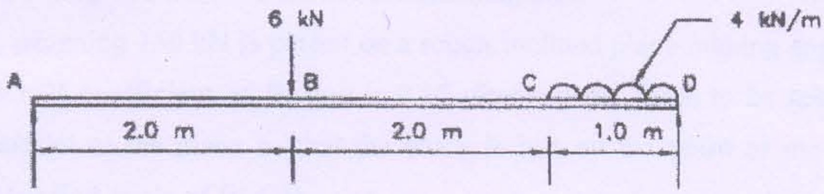

Fig. 7

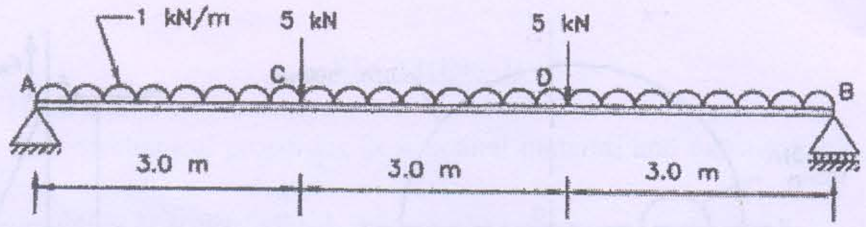

Fig. 8

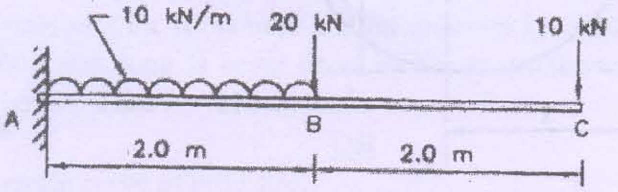

Fig. 9

****